Assignment 1 — Bestselling Books

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Amazon's yearly Top-50 bestsellers, 2009–2022: 700 rows, 7 columns (Name, Author, User Rating, Reviews, Price, Year, Genre). No missing values or duplicate rows, exactly 50 books per year. All results are reproducible (`SEED = 42`).

Task 1 — Basic Python

A list of 7 books (first, last, length, loop) and a price dictionary: average price $12.86, most expensive book found with `max(sample_prices, key=sample_prices.get)` → *Sapiens*.

Task 2 — Load, inspect and clean

Numeric: User Rating, Reviews, Price, Year. Categorical: Name, Author, Genre. 441 unique titles and 305 authors; Non Fiction 388 rows, Fiction 312. First rows of the data:

Name	Author	User Rating	Reviews	Price	Year	Genre
Act Like a Lady, Think Like a...	Steve Harvey	4.6	5013	17	2009	Non Fiction
Arguing with Idiots: How to S...	Glenn Beck	4.6	798	5	2009	Non Fiction
Breaking Dawn (The Twilight S...	Stephenie Meyer	4.6	9769	13	2009	Fiction
Crazy Love: Overwhelmed by a ...	Francis Chan	4.7	1542	14	2009	Non Fiction
Dead And Gone: A Sookie Stack...	Charlaine Harris	4.6	1541	4	2009	Fiction

While inspecting I found a few issues and made these decisions:

Finding	Decision
A row is a book *in a year* — 122 books appear more than once.	`books` (700 rows) for analysis by year; `books_unique` (one row per book, its first year) for books and authors.
Reviews is a snapshot: 97 of those 122 books have the same review count every year.	Don't sum reviews over rows (same reviews counted several times).
One title has hidden trailing spaces and is counted as 2 books.	Strip text columns → **440** unique books.
12 rows with Price = 0 (e.g. *To Kill a Mockingbird*: $0 in 2013–16, $7 in 2019).	Probably promo/free editions → `Price_clean` with NaN, used only for price.

Task 3 — Filtering

Question	Result
Rating ≥ 4.8	148 books
Reviews > 50,000	28 books
cheap_books (Price ≤ 10)	180 books
Avg reviews: rating ≥ 4.8 vs rating < 4.0	15,435 vs 22,355 (only 5 books)

First rows of cheap_books:

Name	Author	Price	Genre
Arguing with Idiots: How to Stop Small Minds and Bi...	Glenn Beck	5	Non Fiction
Dead And Gone: A Sookie Stackhouse Novel (Sookie St...	Charlaine Harris	4	Fiction
Divine Soul Mind Body Healing and Transmission Syst...	Zhi Gang Sha	6	Non Fiction
Eat This Not That! Supermarket Survival Guide: The ...	David Zinczenko	1	Non Fiction
Eclipse (Twilight Sagas)	Stephenie Meyer	7	Fiction

the 5 low-rated books have *more* reviews on average. They're famous, hyped books (Fifty Shades of Grey, The Casual Vacancy, Go Set a Watchman) — controversial books make more people read them and want to give their opinion. Only 5 books though, so not a strong conclusion.

Task 4 — Grouping and features

Genre	Avg rating	Avg price	Avg reviews
Fiction	4.63	$12.46	19,222
Non Fiction	4.62	$13.83	11,538

`year_genre_stats` — average rating and price per year and genre:

Year	Rating Fiction	Rating Non Fiction	Price Fiction	Price Non Fiction
2009	4.59	4.58	$15.58	$15.23
2010	4.62	4.52	$10.21	$16.00
2011	4.62	4.51	$12.20	$17.62
2012	4.50	4.56	$12.29	$17.48
2013	4.55	4.56	$11.68	$18.19
2014	4.63	4.61	$11.80	$20.81
2015	4.65	4.65	$9.94	$10.97
2016	4.72	4.65	$13.33	$13.97
2017	4.74	4.59	$9.22	$13.73
2018	4.74	4.62	$8.76	$11.79
2019	4.82	4.69	$9.35	$10.57
2020	4.73	4.72	$9.05	$11.48
2021	4.74	4.74	$9.16	$12.40
2022	4.69	4.70	$11.69	$13.29

Top by unique books	Top by total reviews	Top by rows (wrong)
Jeff Kinney 14 Rick Riordan 10 Stephenie Meyer 8 Dav Pilkey 8 Suzanne Collins 6	Colleen Hoover 255,531 Suzanne Collins 174,293 Kristin Hannah 146,098 Matt Haig 144,313 E L James 130,746	Jeff Kinney 14 Gary Chapman 14 Suzanne Collins 12 APA 11 Rick Riordan 11

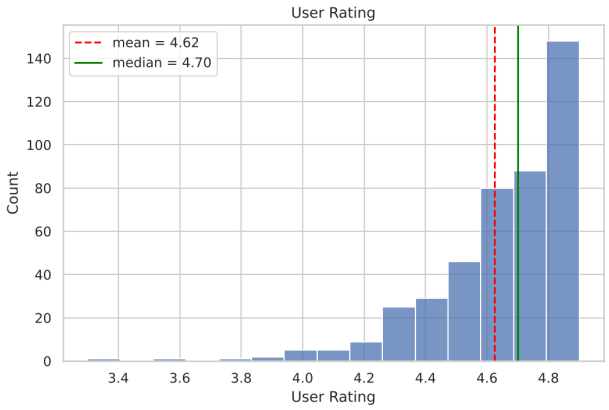
Counting rows lifts authors with one long-running book (Gary Chapman, the APA manual), so I count unique books.

reviews_per_year: the data goes to 2022, so `2020 − Year` gives 0 or negative values. I used `CURRENT_YEAR = 2023` (last year + 1) and the first year on the list, with `clip(lower=1)` against division by 0.

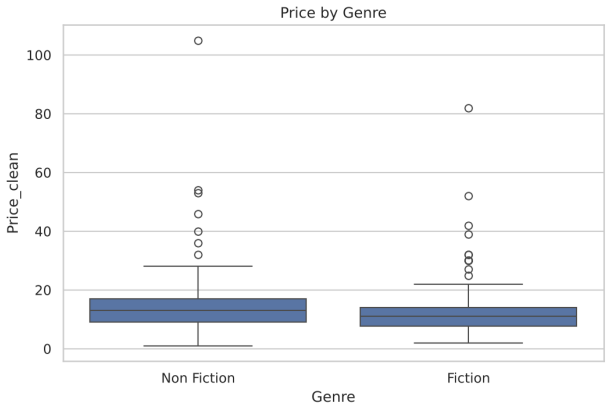
Name	Author	Year	Reviews	reviews_per_year
The Midnight Library: A Novel	Matt Haig	2021	144,313	72,156
Verity	Colleen Hoover	2022	71,826	71,826
Reminders of Him: A Novel	Colleen Hoover	2022	56,049	56,049
The Four Winds: A Novel	Kristin Hannah	2021	96,810	48,405
A Promised Land	Barack Obama	2020	121,109	40,370

raw `Reviews` grows the longer a book is around. `reviews_per_year` shows how *fast* it gets reviews — the top is recent hits (2020–2022).

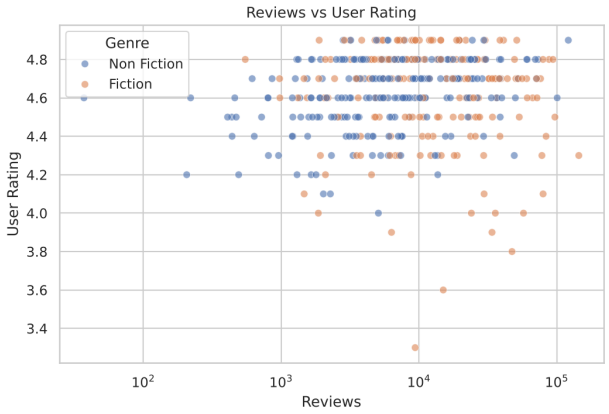
Task 5 — Visualizations



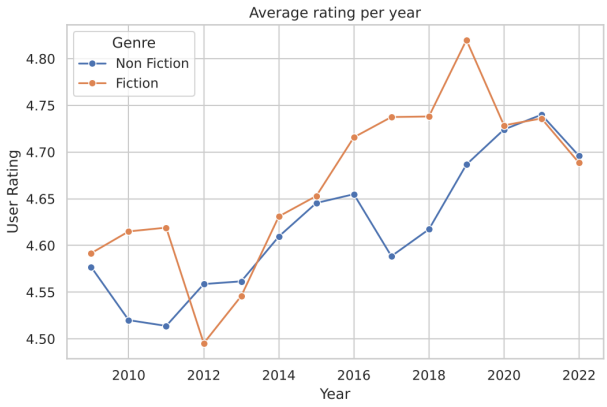
5.1 most ratings are 4.5–4.9. A few low ones pull the mean (4.62) below the median (4.70).



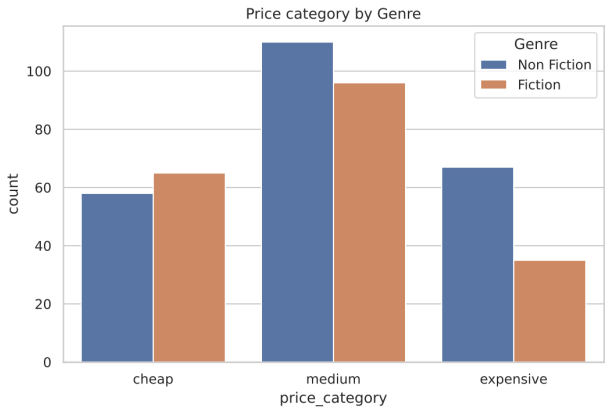
5.2 Non Fiction is a bit more expensive (median 13 vs 11). The very expensive ones are professional manuals (DSM-5 $105, APA manual) and box sets (Twilight collection $82).



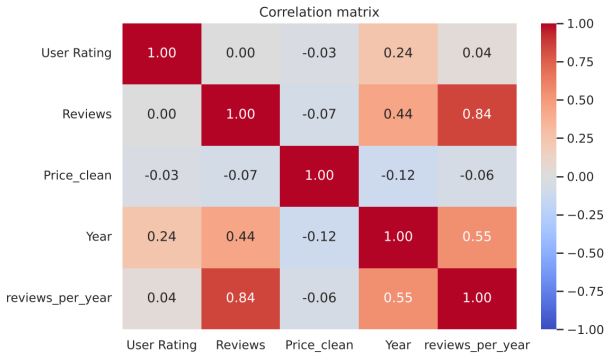
5.3 no clear relation — books with lots of reviews have all kinds of ratings. Log scale because reviews are very skewed.



5.4 rating goes up over time (~4.55 → ~4.73) in both genres. I think it's rating inflation (people give more stars), not better books.



5.5 cheap ≤ $8, medium $9–16, expensive > $16 (25%/75% quantiles). mostly medium in both. Non Fiction has about twice as many expensive books (67 vs 35) — the manuals and guides.



6.1 Correlation matrix (unique books, incl. reviews_per_year).

Price statistics by genre (5.2) and price category counts (5.5):

Price	Fiction	Non Fiction
count	196.0	235.0
mean	12.5	13.8
std	8.6	9.6
min	2.0	1.0
median	11.0	13.0
max	82.0	105.0

Category	Fiction	Non Fiction
cheap	65	58
medium	96	110
expensive	35	67

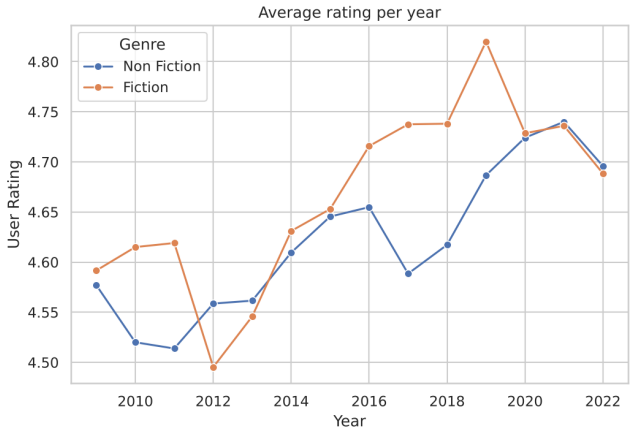
Task 6 — Correlations and blockbusters

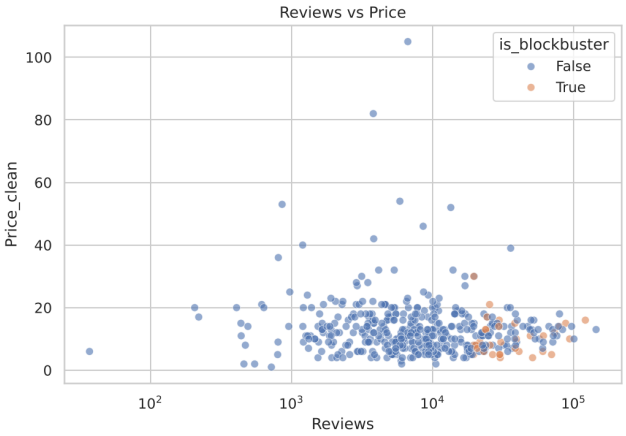
• Reviews vs Rating ≈ 0 — popular doesn't mean liked
• Price vs Rating ≈ 0 — price doesn't affect rating
• Year vs Reviews 0.44, Year vs Rating 0.24 — newer books get more reviews and slightly higher ratings (more users + rating inflation)

Blockbuster rule: rating ≥ 4.8 **and** reviews ≥ 75th percentile (19,019). 4.7 is the median rating so it barely filters; the reviews threshold comes from the data instead of a random number.

36 blockbusters (8.2%). No author has more than 2 and the genres are balanced, so success comes from many different good books, not a few star authors. 17 are from 2020 — lots of kids' books and workbooks (COVID, kids at home). They're also cheaper (median $9 vs $12).

Task 7 — Summary





Almost every bestseller is rated 4.5–4.9, so rating is the minimum to get on the list but doesn't separate the books — the number of reviews varies much more and shows popularity better. Rating and reviews aren't correlated at all; controversial books even get more reviews. Ratings also rise over the years, which looks like rating inflation.

Non Fiction is a bit more expensive than Fiction (avg ~$13.8 vs ~$12.5), mainly because of professional manuals and guides, but the gap is small.

No author or genre dominates the blockbusters — they come from many different authors. Many are recent and cheap, especially kids' books from 2020 (COVID).

What I'd tell a publisher: a good rating is a must, but what bestsellers really have in common is getting a lot of reviews fast, an affordable price (~$10), and a wide audience — e.g. kids and families.